

SHIVAM MAURYA

Robotics Software Engineer

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TECHNICAL SKILLS

Languages: C++17 · Python 3 · Bash · CMake · SQL

Robotics Fundamentals: Kinematics · Dynamics · Control Systems · State Estimation · Sensor Fusion · Joint Control

ROS / ROS 2: rclcpp · rclpy · Topics/Services/Actions/Parameters · Lifecycle & Component Nodes · Custom Interfaces (.msg/.srv/.action) · colcon · ament_cmake · rosddep · tf2

Robot Control & Motion: ros2_control · hardware_interface · MoveIt 2 · OMPL (RRT*, RRTConnect, PRM) · Nav2 · Behavior Trees · DWB/Navfn Planners · robot_localization (EKF)

Simulation & Modelling: Gazebo Classic · Ignition Gazebo · RViz2 · URDF/XACRO · SLAM Toolbox · Sim-to-Real Deployment

Perception & Sensors: LiDAR · Depth Camera (RealSense D435i) · IMU · Encoders · Camera Calibration · OpenCV · YOLO · ONNX Runtime · TensorFlow · CNN

Embedded & Communication: Arduino · ESP32 · ESP8266 · NVIDIA Jetson · UART · I2C · SPI · PWM · Motor Drivers (L298N, A4988) · GSM/SIM800L

Tools: Git/GitHub · Docker · Linux (Ubuntu) · VS Code · Jupyter · CI/CD · Postman

EXPERIENCE

Robotics Software Developer — Student Researcher

Jun 2025 – Present

Army Institute of Technology, Pune

- Developed an autonomous **navigation stack** using **ROS 2 Humble**, **Nav2**, and **SLAM Toolbox** on a differential-drive robot, fusing LiDAR and depth-camera data via a **robot_localization EKF** for GPS-denied localization.
- Implemented **MoveIt 2** motion-planning pipelines for a 6-DOF manipulator; configured **OMPL planners (RRT*, RRTConnect, PRM)** for collision-aware, Cartesian pick-and-place execution.
- Authored **ROS 2 nodes in C++ and Python** for sensor processing, hardware interfacing (UART/I2C/SPI), odometry publishing, and actuator control.
- Deployed **YOLO-based real-time perception pipelines** as ROS 2 nodes, achieving **30+ FPS on CPU-only embedded hardware** for obstacle detection and object classification.

Open-Source Contributor — ROS 2 Ecosystem

Jun 2025 – Present

navigation2 · ros2_control · ros2_control_demos · ros2_controllers · ros2/demos · ros2_documentation

- Shipped 15+ merged pull requests** and maintain 5+ open PRs across core ROS 2 repositories; documented real-time-safe methods in ros2_control's **controller_interface** headers and investigated a ResourceManager/ControllerManager initialization-strictness regression tied to a gz_ros2_control crash.
- Fixed CI/build issues across repositories (ROS 2 Killed spawner config migration, headless CI display, SetuptoolsDeprecationWarning) and added **ament_mypy** static-analysis linting to core codebases.

Software Lead (previously Joint Secretary) — Centre of Excellence for AI & Robotics

Jun 2024 – Present

Army Institute of Technology, Pune

- Co-lead a **100+ member** AI & Robotics research cell; organize robotics workshops, autonomous-systems hackathons, and ROS 2 project sprints.
- Represented AIT at **ABU Robocon 2025/26**; led team technical strategy including system architecture, CNN-based real-time vision pipeline deployment, and competition-robot integration.

PROJECTS

Autonomous Ground Vehicle — Nav2 Full-Stack Navigation

ROS 2 Humble, Nav2, SLAM Toolbox, LiDAR, C++, Python

- Deployed a full **Nav2 navigation stack** (global/local costmaps, DWB local planner, recovery Behavior Trees) on a differential-drive robot, with SLAM Toolbox for real-time 2D LiDAR mapping and a robot_localization EKF for GPS-denied localization.

6-DOF Robotic Arm — AI-Guided Pick & Place

ROS 2, MoveIt 2, URDF/XACRO, OMPL, ros2_control

- Built a **MoveIt 2** pipeline (OMPL RRT*/RRTConnect/PRM) for collision-aware trajectory planning, with **CNN-based grasp-pose estimation** from depth-camera input driving approach → grasp → lift → place execution via ros2_control on physical hardware.

Real-Time Perception Pipeline — ROS 2 Vision Node

ROS 2, YOLO, TensorFlow, ONNX, OpenCV

- Built a modular perception node with hot-swappable YOLO/CNN backends exported to **ONNX** for **30+ FPS CPU-only inference**; validated live in ABU Robocon competition, driving robot decision-making directly.

Differential-Drive Robot — ROS 2 Fundamentals

ROS 2 Humble, C++, Python, URDF, RViz2, Gazebo, tf2

- Built a two-wheeled robot from scratch, authoring URDF with accurate wheel geometry, joint definitions, and collision meshes.
- Implemented custom C++ nodes for `cmd_vel` subscription, encoder-based odometry computation, and tf2 transform broadcasting (`base_link` → `odom`); simulated in Gazebo and visualized live in RViz2.

CNN-Based Vision Classifier — Robocon Competition Robot

ROS 2, Python, YOLO, OpenCV, TensorFlow

- Built and trained a YOLO-based object detection and classification model on a custom-labeled dataset for the ABU Robocon competition robot.
- Integrated the vision pipeline into the ROS 2 architecture as a dedicated perception node publishing detections on a custom message topic; achieved **30+ FPS** on embedded hardware and deployed live on the competition robot.

NanoTracker-24/7 — Ultra-Low-Power Asset Tracker

C++, Arduino, ESP8266, SIM800L GSM, LBS Triangulation, EEPROM

- Engineered firmware with a **99% deep-sleep duty cycle**; implemented GPS-free LBS cell-tower triangulation (LAC+CID) for location tracking that works indoors and inside metal containers.
- Implemented dual SMS telemetry, ADC battery-voltage monitoring, watchdog fail-safe recovery, and OTA SMS configuration persisted in EEPROM.

EDUCATION

B.E. Automation and Robotics Engineering — Army Institute of Technology, Pune

Jun 2024 – May 2028

CGPA: 8.0 / 10

Relevant Coursework: Robotics Systems · Control Theory · Embedded Systems · Computer Vision · Machine Learning · IoT & Sensor Networks · Signals & Systems

ACHIEVEMENTS & EXTRACURRICULARS

- **ABU Robocon 2025/26** — Led software architecture and real-time CNN-based vision deployment on the competition robot; represented AIT Centre of Excellence for AI and Robotics.
- **Open-Source Contributor** — 15+ merged pull requests, 5+ open, across core ROS 2 repositories (`navigation2`, `ros2_control`, `ros2_control_demos`, `ros2_controllers`, `ros2/demos`, `ros2_documentation`).
- **IIIT Pune Thinking Machines 2025** — MedSynapse multi-disease AI platform selected for the Diagnostic Tools track (4 ML/DL models, deployed on Streamlit Cloud).
- **CODECURE Hackathon 2025 (Track B), IIT BHU** — Top submission; antimicrobial-resistance predictor at 99.27% accuracy, WHO/CLSI-aligned.
- **Embedded Systems** — Built 40+ Arduino/ESP32 embedded projects spanning sensor systems, motor control, IoT communication, and automation.